

Every Knot Admits a Construction Word: Proof

A Proof from Rational Density

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Definitions

The **stick number** $s(K)$ of a knot K is the minimum number of straight-line segments required to form a polygonal realization of K in $\mathbb{R}^3$.

A **construction word** for a knot is a finite sequence of sphere-intersection operations in three-dimensional Euclidean space, beginning from three initial points (the base triangle), where each step intersects three spheres centered at previously constructed points, producing exactly two candidate vertices (target and mirror). The sequence of binary choices (target or mirror at each step) that yields a given knot type constitutes the construction word for that knot.

Theorem (Rational Realizability)

For every tame knot K and every integer $n \geq s(K)$, there exists a polygonal realization of K with n edges whose vertices all have rational coordinates.

Proof

The argument requires only three ingredients: that non-touching things have space between them, that the space between them is enough to adjust the vertices, and that you can always adjust by a rational amount. No infinitesimal or zero-thickness idealization is needed.

1. Non-adjacent edges that don't touch have positive separation.

Take any valid polygonal realization of K with n edges. The edges are line segments of positive length. In any valid realization, non-adjacent edges do not intersect — they miss each other entirely. Because we have finitely many edges (and therefore finitely many pairs of non-adjacent edges), and each such pair consists of two compact sets that are disjoint, each pair has a strictly positive minimum distance between them. Let $\delta > 0$ be the smallest such distance across all non-adjacent edge pairs.

This δ is a concrete, finite, positive number determined by the specific configuration. It is the amount of room available in the configuration.

Note that this holds regardless of whether we model the edges as idealized zero-thickness line segments or as physical tubes of some small positive radius. If the edges are tubes of radius $r > 0$, the non-intersection condition means the tube surfaces don't touch, which means the central line segments are separated by at least $2r$, and $\delta \geq 2r > 0$. The argument is the same either way: things

that don't touch have space between them.

2. Small adjustments to the vertices preserve the knot type.

If we perturb each vertex by a distance less than $\delta/2$, then each edge moves by less than $\delta/2$ (since an edge is determined by its two endpoints). Any pair of non-adjacent edges that was separated by distance δ is now separated by at least $\delta - 2 \cdot (\delta/2) = 0$, but in fact strictly more than zero because the worst case requires both edges to move their closest points directly toward each other, which generically does not happen. More carefully: a perturbation of less than $\delta/4$ per vertex guarantees that each edge moves by less than $\delta/4$, so any pair of non-adjacent edges remains separated by at least $\delta - 2 \cdot (\delta/4) = \delta/2 > 0$.

Since no edge crossing is created or destroyed — non-adjacent edges that were separated remain separated, and adjacent edges still share their common vertex — the knot type is preserved. This is a standard result in geometric knot theory (Fact 2.2.4 in Roberts, *Knot Knots*; Lemma 2.6.1 in Buck and Simon, 1993): sufficiently small perturbations of the vertices of a polygonal knot preserve the knot type.

The key point is that "sufficiently small" is a finite positive bound ($\delta/4$), not an infinitesimal. There is a concrete amount by which you can adjust each vertex without changing the knot.

3. You can always adjust by a rational amount.

The rationals $\mathbb{Q}$ are dense in $\mathbb{R}$: between any two real numbers lies a rational number. Consequently, within any ball of positive radius around any point in $\mathbb{R}^3$, there is a point with all-rational coordinates. In particular, within a ball of radius $\delta/4$ around each vertex of our realization, there is a point with rational coordinates.

Replacing each vertex with such a nearby rational point perturbs each vertex by less than $\delta/4$, which by step 2 preserves the knot type.

Therefore there exists a realization of K with n edges whose vertex coordinates are all rational. $\square$

Corollary (Construction Word Existence)

Every tame knot admits a construction word at every stick number at or above its minimum.

Proof. Every rational number is constructible (reachable from integers by arithmetic operations, which are a subset of compass-and-straightedge operations). Every constructible point in $\mathbb{R}^3$ is, by definition, the output of a finite sequence of sphere-and-plane intersections starting from rational data. Each such intersection produces exactly two candidate points (the two roots of the resulting quadratic equation), and the constructible point is obtained by selecting one at each step.

Given a rational-coordinate realization of K (which exists by the theorem), we can order the vertices and, for each vertex after an initial base triangle, identify a sphere-intersection sequence that produces it. The sequence of binary selections (target or mirror at each step) constitutes a construction word for K .

Since the theorem provides a rational realization at every stick number $n \geq s(K)$, a construction word exists at every such n . This universality across stick numbers is essential: it means any

invariant defined via the construction word can be computed at any polygonal resolution at or above the minimum, which is what makes such an invariant well-defined independent of stick number. $\square$

Remarks

On the character of the proof. The proof uses only the Archimedean property of the real numbers (between any two reals lies a rational) and the finiteness of the edge configuration (finitely many edges means the minimum pairwise separation is achieved and is positive). It does not require any passage to a limit, any zero-thickness idealization, or any completed infinity. The edges can be understood as physical objects with positive thickness — however small — and the argument goes through unchanged, because the only requirement is that things which don't touch have a finite positive distance between them. This is a concrete physical intuition, not an abstract mathematical idealization.

On computability. The rational realizability theorem says that every knot at every stick number can be realized with vertex coordinates that are ratios of integers. All sphere radii-squared in the corresponding construction are therefore rational. The entire sphere-intersection construction can be carried out in exact rational arithmetic with no floating-point error, and the distribution of knot types across the 2^n sphere-intersection alternatives can in principle be computed exactly.

On non-uniqueness. The construction word for a given knot is far from unique. The same knot at the same stick number admits many different construction words: different choices of base triangle, different orderings of the constructed vertices, and different sphere-intersection sequences reaching the same vertex all yield distinct words. The proof establishes existence of at least one construction word for every knot at every stick number, but does not address the relationship between different construction words for the same knot.

On the Euclidean field. The rational numbers are a subfield of the Euclidean field (the field of compass-and-straightedge constructible numbers, equivalently the maximal tower of quadratic extensions over $\mathbb{Q}$). The Euclidean field is the natural algebraic setting for the construction word framework, since it is precisely the closure of $\mathbb{Q}$ under the sphere-intersection operation. The rational realizability theorem is stronger than a Euclidean field realizability theorem (since $\mathbb{Q} \subset \mathbb{E}$), but the Euclidean field provides the correct conceptual framing: the construction word is a sequence of sphere intersections, and the Euclidean field is the set of numbers those intersections can produce. That every knot can be realized within the much smaller subfield $\mathbb{Q}$ is a computational bonus, not a conceptual necessity.

An Open Direction: The Decoherence Ratio

Consider a construction word for a knot K with n constructed vertices (vertices beyond the initial base triangle). Each constructed vertex has two candidate positions — target and mirror — so there are 2^n total alternative polygons obtained by independently choosing target or mirror at each vertex. Some of these alternatives will realize the same knot type K . Others will realize simpler knots or the unknot.

The fraction of alternatives that preserve K could be called the **decoherence ratio** of the construction, by analogy with quantum decoherence: the knot "survives" only when all the binary

choices remain coherent with each other, and "decoheres" into something simpler when they don't.

This ratio is interesting to explore because it appears, experimentally, to be a knot invariant — independent of the choice of realization, base triangle, and stick number. Computational experiments across multiple random realizations, base triangle choices, and stick numbers for the trefoil (3_1), figure-eight (4_1), and cinquefoil (5_1) yield the following:

Knot	Crossing number c	Observed ratio
Trefoil	3	$2/2^3 = 1/4$
Figure-eight	4	$2/2^4 = 1/8$
Cinquefoil	5	$2/2^5 = 1/16$

The pattern suggests a general formula: the decoherence ratio equals $2/2^c$, where c is the crossing number. If this holds, each crossing imposes one independent binary constraint on the construction, halving the space of valid alternatives, with one degree of freedom (the global target/mirror flip) always remaining free.

This would give the crossing number a direct geometric interpretation: it is the number of independent binary constraints that three-dimensional sphere-intersection geometry imposes on the construction of a knot. It would also provide a method of computing crossing number purely from the construction word, with no reference to knot diagrams, projections, or Reidemeister moves.

This has not yet been proven. The experimental evidence is consistent across all tested cases, including non-degenerate realizations at multiple stick numbers, but a proof that the ratio is invariant across all construction words for a given knot — and that the formula $2/2^c$ holds generally — remains open.

References

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